Reg. No. | | | | | | | | | | |

[18AS35]

THIRD SEMESTER B.Tech EXAMINATION, DECEMBER–2019

ENGINEERING MECHANICS

Time:3 Hours **Maximum Marks:70**

Instructions: i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carry 14 marks.

ii. Missing data may be suitably assumed.

1. a. With a neat sketch explain stress v/s strain diagram for ductile and brittle materials **6**

b. A bar of cross section 10mmX10mm is subjected to an axial pull of 8000N. The lateral dimension of the bar is found to be changed to 9.9985mmX9.9985mm. if the modulus of rigidity is 0.8×10^5 N/mm^2, determine the Poisson's ratio and modulus of elasticity. **8**

2. a. Derive equation for stresses on inclined planes. **6**

b. A rod of 2m long at a temperature of 10° C. Find the expansion of the rod when the temperature is raised to 80°C. If the expansion is prevented, find the stress induced in the material of the rod. Take E=1 X10^5Mpa and $\propto = 12X10^{-6}/^{o}c$. **8**

3. a. Define: **08**

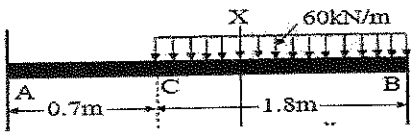
i. Neutral axis

ii. Section modulus

iii. Flexural rigidity

iv. Moment of resistance

b. A cantilever beam caries a Uniformly distributed load as shown in figure. **06**

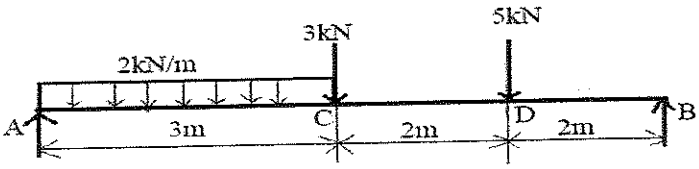


Draw Shear force diagram and Bending moment diagram for the beam.

OR

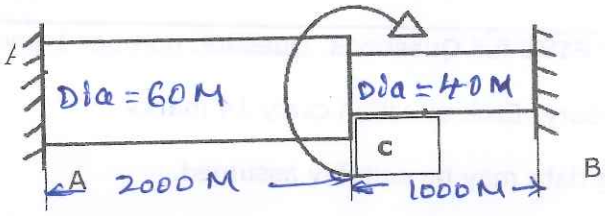
4. a. A Cantilever beam of length 2 m fails when a load of 2 kN is applied at free end. If the section is 40 mm×60 mm. Find the stress at failure. **06**

b. For the beam as shown, calculate shear force and bending moments at salient points and draw shear force and bending moment diagrams. **08**



5. a. Two solid shafts of Aluminium and steel as shown in fig is subjected to a torque of 300 Nm applied at C. Calculate the shearing stress in each of the material and angle of twist at junction and torque reactions at A and B. **10**

$G_{AL}=3\times10^4$ N/mm^2 ,G_{steel}= 9×10^4N/mm^2



b. Explain Moment of inertia and polar moment of inertia. **04**

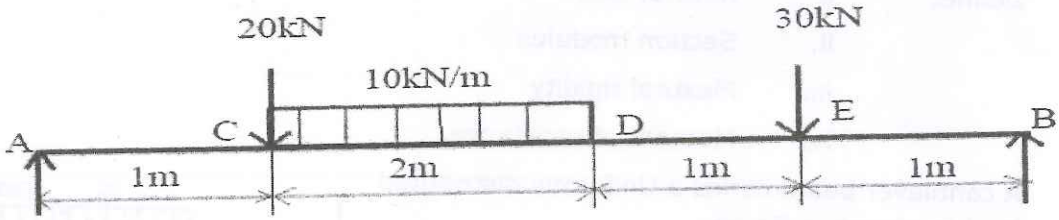
OR

6. a. Find maximum power that can be transmitted through a 50 mm diameter shaft at 150 rpm, if permissible shear stress is 80 N/mm^2. **06**

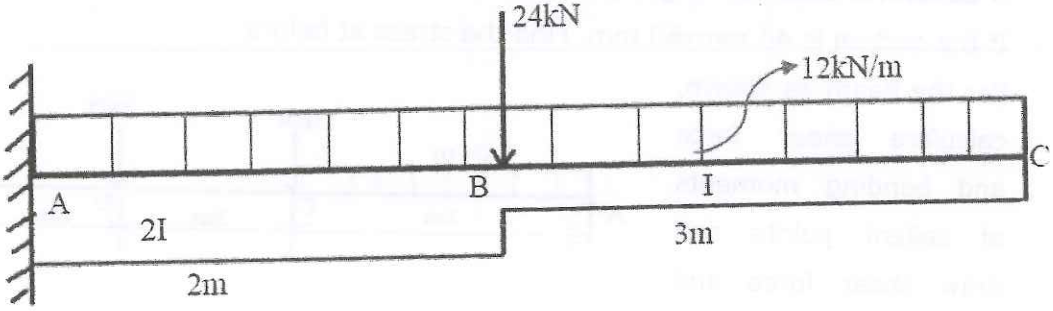
b. Design the diameter of solid circular shaft to transmit 50 KW power at 100 rpm. Max torque is likely to exceed mean torque by 25%. Maximum permissible shear stress is 60 MPa. Also calculate angle of twist for a length of 2m. Take G=80GPa **08**

7. a. Explain the concept of slope and deflection for a simply supported beam with boundary conditions. **06**

b. Determine the deflection at points B, C and D in the beam shown below. The beam has circular cross section of 200 mm diameter. Take E=200 GPa **08**



OR

8 a. Explain the applications of energy theorems for deflection in beams. **04**

b. Find slope and deflection at free end by conjugate beam method: **10**